UNIT 1 · KINEMATICS · BEFORE THE CART LAB · HONORS

Mini Quiz (Version 2) — Kinematics Readiness

Physics 2026–27 · 20 minutes · 20 points · in class Tuesday, October 6

NAME _____ DATE _____

Three problems, about 20 minutes. These are Problem Set 5 (Version 2) problems with new numbers: run KDEA on each, and name the *missing variable* in your E row — a right answer with no E row does not earn full credit. The four equations are printed below; choosing between them is the quiz. One given arrives in km/h — convert first. No notes; a calculator is fine. Units the whole way.

$v_f = v_i + at$ · $\Delta x = v_i t + \frac{1}{2}at^2$ · $v_f^2 = v_i^2 + 2a\Delta x$ · $\Delta x = \frac{1}{2}(v_i + v_f)t$

1 The cart on the ramp

KDEA · 7 POINTS

A lab cart is released from rest at the top of a ramp and rolls with constant acceleration. It passes the $0.90\ m$ mark $1.5\ s$ after release.

a. Find the cart's acceleration.

b. How fast is it moving as it passes that mark?

c. Where was the cart at $t = 0.50\ s$ — one-third of the time? Answer with a proportionality argument, not a fresh KDEA.

d. If you plotted x vs t^2 for this cart, what would the slope be, with units?

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

2 No time given

KDEA · 7 POINTS

A car moving at 72 km/h brakes with constant acceleration and stops in 50 m.

a. Find the acceleration, sign included.

b. Find the stopping time two ways: with the kinematics equations, and with $t = \Delta x / v_{avg}$.

c. How fast is the car moving at the halfway point, 25 m into the stop? Explain in one sentence why it is *not* half the starting speed.

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3 Three-stage trip

MULTI-STEP · 6 POINTS

A tram starts from rest, accelerates at 1.5 m/s² for 8.0 s, holds that velocity for 10 s, then brakes uniformly to rest over 36 m.

a. How fast is it moving at the end of the first stage, and how far has it gone?

b. Find its acceleration during braking (sign included) and how long the braking takes.

c. Find the total distance and the average velocity for the whole trip.

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Formative check — you may retake a similar version once (syllabus quiz policy). The cart lab starts right after this; Problem 1 is the experiment you are about to run.